

ALFAAH58023

## 3-(Ethoxycarbonyl)propylzinc bromide, 0.5M in THF

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 3-(乙氧基氨基甲酰)丙基锌溴化物, 四氢呋喃溶液  
 3-(Ethoxycarbonyl)propylzinc bromide, 0.5M in THF

**Cat No. :** H58023  
**Molecular Formula** C6 H11 BrO2 Zn

**Supplier** Avocado Research Chemicals Ltd.  
 (Part of Thermo Fisher Scientific)  
 Shore Road, Heysham  
 Lancashire, LA3 2XY,  
 United Kingdom  
 Office Tel: +44 (0) 1524 850506  
 Office Fax: +44 (0) 1524 850608

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
 Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
Yellow - Brown - Black

**Odor**  
No information available

#### Emergency Overview

Highly flammable liquid and vapor. Suspected of causing cancer. May cause respiratory irritation. In contact with water releases flammable gases which may ignite spontaneously. Harmful if swallowed. Causes severe skin burns and eye damage. May cause drowsiness and dizziness. May form explosive peroxides. Air sensitive.

#### Classification of the substance or mixture

|                                                                        |              |
|------------------------------------------------------------------------|--------------|
| Flammable liquids.                                                     | Category 2   |
| Substances/mixtures which, in contact with water, emit flammable gases | Category 1   |
| Acute Oral Toxicity                                                    | Category 4   |
| Skin Corrosion/Irritation                                              | Category 1 B |
| Serious Eye Damage/Eye Irritation                                      | Category 1   |
| Carcinogenicity                                                        | Category 2   |
| Specific target organ toxicity - (single exposure)                     | Category 3   |

#### Label Elements

## 3-(Ethoxycarbonyl)propylzinc bromide, 0.5M in THF



## Signal Word

Danger

## Hazard Statements

H225 - Highly flammable liquid and vapor  
H260 - In contact with water releases flammable gases which may ignite spontaneously  
H351 - Suspected of causing cancer  
H335 - May cause respiratory irritation  
H302 - Harmful if swallowed  
H314 - Causes severe skin burns and eye damage  
H336 - May cause drowsiness or dizziness

## Precautionary Statements

## Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P231 + P232 - Handle and store contents under inert gas. Protect from moisture  
P240 - Ground and bond container and receiving equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

## Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P302 + P335 + P334 - IF ON SKIN: Brush off loose particles from skin. Immerse in cool water  
P362 + P364 - Take off contaminated clothing and wash it before reuse

## Storage

P402 + P404 - Store in a dry place. Store in a closed container

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Physical and Chemical Hazards

Vapors may cause flash fire or explosion. Highly flammable. Reacts violently with water, liberating extremely flammable gases. May form explosive peroxides.

## Health Hazards

Suspected of causing cancer. May cause respiratory irritation. Harmful if swallowed. Corrosive. Causes skin and eye burns. May cause drowsiness or dizziness.

## Environmental hazards

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Is not likely mobile in the environment due its low water solubility. Spillage unlikely to penetrate soil. The product is insoluble and floats on water.

Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

# SAFETY DATA SHEET

## 3-(Ethoxycarbonyl)propylzinc bromide, 0.5M in THF

| Component                            | CAS No      | Weight % |
|--------------------------------------|-------------|----------|
| Tetrahydrofuran                      | 109-99-9    | 87       |
| 3-(Ethoxycarbonyl)propylzinc bromide | 131379-39-0 | 13       |

### SECTION 4. FIRST AID MEASURES

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.

#### Inhalation

If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.

#### Ingestion

Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.

#### Most important symptoms and effects

Causes burns by all exposure routes. Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

#### Notes to Physician

Treat symptomatically. Symptoms may be delayed.

### SECTION 5. FIRE-FIGHTING MEASURES

#### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

No information available.

#### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### SECTION 6. ACCIDENTAL RELEASE MEASURES

# SAFETY DATA SHEET

## 3-(Ethoxycarbonyl)propylzinc bromide, 0.5M in THF

### Personal Precautions

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Should not be released into the environment. See Section 12 for additional Ecological Information. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system.

### Methods for Containment and Clean Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7. HANDLING AND STORAGE

### Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. If peroxide formation is suspected, do not open or move container. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

### Storage

Keep refrigerated. Corrosives area. Keep containers tightly closed in a dry, cool and well-ventilated place. Containers should be dated when opened and tested periodically for the presence of peroxides. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals. Keep away from heat, sparks and flame.

### Specific Use(s)

Use in laboratories

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component       | China                      | Taiwan                                     | Thailand     | Hong Kong                                                                                  |
|-----------------|----------------------------|--------------------------------------------|--------------|--------------------------------------------------------------------------------------------|
| Tetrahydrofuran | TWA: 300 mg/m <sup>3</sup> | TWA: 200 ppm<br>TWA: 590 mg/m <sup>3</sup> | TWA: 200 ppm | TWA: 200 ppm<br>TWA: 590 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 737 mg/m <sup>3</sup> |

| Component       | ACGIH TLV                            | OSHA PEL                                                                                                                                                                         | NIOSH                                                                                                        | The United Kingdom                                                                                                        | European Union                                                                                                              |
|-----------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | TWA: 50 ppm<br>STEL: 100 ppm<br>Skin | (Vacated) TWA: 200 ppm<br>(Vacated) TWA: 590 mg/m <sup>3</sup><br>(Vacated) STEL: 250 ppm<br>(Vacated) STEL: 735 mg/m <sup>3</sup><br>TWA: 200 ppm<br>TWA: 590 mg/m <sup>3</sup> | IDLH: 2000 ppm<br>TWA: 200 ppm<br>TWA: 590 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 735 mg/m <sup>3</sup> | STEL: 100 ppm 15 min<br>STEL: 300 mg/m <sup>3</sup> 15 min<br>TWA: 50 ppm 8 hr<br>TWA: 150 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 50 ppm (8h)<br>TWA: 150 mg/m <sup>3</sup> (8h)<br>STEL: 100 ppm (15min)<br>STEL: 300 mg/m <sup>3</sup> (15min)<br>Skin |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists  
OSHA - Occupational Safety and Health Administration  
NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Monitoring methods

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BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography MDHS 91 Metals and metalloids in workplace air by X-ray fluorescence spectrometry MDHS 99 Metals in air by ICP-AES

**Exposure Controls****Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material  | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|-----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber  | See manufacturers | -               | EN 374      | (minimum requirement) |
| Viton (R)       | recommendations   |                 |             |                       |
| Butyl rubber    |                   |                 |             |                       |
| Neoprene gloves |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown conforming to EN14387

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**

**Appearance** Yellow - Brown - Black  
**Physical State** Liquid

**Odor** No information available

## SAFETY DATA SHEET

## 3-(Ethoxycarbonyl)propylzinc bromide, 0.5M in THF

|                                                |                          |                                             |
|------------------------------------------------|--------------------------|---------------------------------------------|
| <b>Odor Threshold</b>                          | No data available        |                                             |
| <b>pH</b>                                      | No information available |                                             |
| <b>Melting Point/Range</b>                     | No data available        |                                             |
| <b>Softening Point</b>                         | No data available        |                                             |
| <b>Boiling Point/Range</b>                     | No information available |                                             |
| <b>Flash Point</b>                             | No information available | <b>Method</b> - No information available    |
| <b>Evaporation Rate</b>                        | No data available        |                                             |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                      |
| <b>Explosion Limits</b>                        | No data available        |                                             |
| <b>Vapor Pressure</b>                          | No data available        |                                             |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.976 g/cm3              | @ 20 °C                                     |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                      |
| <b>Water Solubility</b>                        | Immiscible               |                                             |
| <b>Solubility in other solvents</b>            | No information available |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                             |
| <b>Component</b>                               | <b>log Pow</b>           |                                             |
| Tetrahydrofuran                                | 0.45                     |                                             |
| <b>Autoignition Temperature</b>                | No data available        |                                             |
| <b>Decomposition Temperature</b>               | No data available        |                                             |
| <b>Viscosity</b>                               | No data available        |                                             |
| <b>Explosive Properties</b>                    |                          | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | No information available |                                             |
| <b>Molecular Formula</b>                       | C6 H11 BrO2 Zn           |                                             |
| <b>Molecular Weight</b>                        | 260.44                   |                                             |

## SECTION 10. STABILITY AND REACTIVITY

|                                 |                                                                   |
|---------------------------------|-------------------------------------------------------------------|
| <b>Stability</b>                | Air sensitive.                                                    |
| <b>Hazardous Reactions</b>      | None under normal processing.                                     |
| <b>Hazardous Polymerization</b> | No information available.                                         |
| <b>Conditions to Avoid</b>      | Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Materials to avoid</b>       | Strong bases.                                                     |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen bromide. Metal oxides.

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

(a) acute toxicity;  
Toxicology data for the components

| Component       | LD50 Oral          | LD50 Dermal           | LC50 Inhalation                               |
|-----------------|--------------------|-----------------------|-----------------------------------------------|
| Tetrahydrofuran | 1650 mg/kg ( Rat ) | > 2000 mg/kg (Rabbit) | 180 mg/L ( Rat ) 1 h<br>53.9 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

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**Respiratory  
Skin**No data available  
No data available

| Component                          | Test method                                       | Test species | Study result    |
|------------------------------------|---------------------------------------------------|--------------|-----------------|
| Tetrahydrofuran<br>109-99-9 ( 87 ) | Local Lymph Node Assay<br>OECD Test Guideline 429 | mouse        | non-sensitising |

**(e) germ cell mutagenicity;**

No data available

| Component                          | Test method                                             | Test species          | Study result |
|------------------------------------|---------------------------------------------------------|-----------------------|--------------|
| Tetrahydrofuran<br>109-99-9 ( 87 ) | OECD Test Guideline 476<br>Gene cell mutation           | in vivo<br>Mammalian  | negative     |
|                                    | OECD Test Guideline 473<br>Chromosomal aberration assay | in vitro<br>Mammalian | negative     |

**(f) carcinogenicity;**

Category 2

Limited evidence of a carcinogenic effect The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component       | EU | UK | Germany | IARC     |
|-----------------|----|----|---------|----------|
| Tetrahydrofuran |    |    |         | Group 2B |

**(g) reproductive toxicity;**

No data available

| Component                          | Test method             | Test species / Duration | Study result      |
|------------------------------------|-------------------------|-------------------------|-------------------|
| Tetrahydrofuran<br>109-99-9 ( 87 ) | OECD Test Guideline 416 | Rat 2 Generation        | NOAEL = 3,000 ppm |

**(h) STOT-single exposure;**

Category 3

**Results / Target organs**Respiratory system  
Central nervous system (CNS)**(i) STOT-repeated exposure;**

No data available

**Target Organs**

No information available.

**(j) aspiration hazard;**

No data available

**Symptoms / effects, both acute and delayed**

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**SECTION 12. ECOLOGICAL INFORMATION****Ecotoxicity effects**

May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component       | Freshwater Fish                                                                        | Water Flea                                   | Freshwater Algae | Microtox |
|-----------------|----------------------------------------------------------------------------------------|----------------------------------------------|------------------|----------|
| Tetrahydrofuran | 2160 mg/l LC50 = 96 h<br>Pimephales promelas<br>Leuciscus idus: LC50:<br>2820 mg/L/48h | EC50 48 h 3485 mg/l<br>EC50: >10000 mg/L/24h |                  |          |

**Persistence and Degradability**

Product contains heavy metals. Discharge into the environment must be avoided. Special

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**Persistence  
Degradation in sewage  
treatment plant**

pre-treatment is necessary  
May persist, based on information available.  
Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential**

May have some potential to bioaccumulate

| Component       | log Pow | Bioconcentration factor (BCF) |
|-----------------|---------|-------------------------------|
| Tetrahydrofuran | 0.45    | No data available             |

**Mobility in soil**

Spillage unlikely to penetrate soil The product is insoluble and floats on water Is not likely mobile in the environment due its low water solubility

**Endocrine Disruptor Information**

| Component       | EU - Endocrine Disruptors<br>Candidate List | EU - Endocrine Disruptors -<br>Evaluated Substances | Japan - Endocrine Disruptor<br>Information |
|-----------------|---------------------------------------------|-----------------------------------------------------|--------------------------------------------|
| Tetrahydrofuran | Group III Chemical                          |                                                     |                                            |

**Persistent Organic Pollutant  
Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused  
Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

UN-No UN3399  
 Proper Shipping Name ORGANOMETALLIC SUBSTANCE, LIQUID, WATER-REACTIVE, FLAMMABLE  
 Technical Shipping Name (3-(Ethoxycarbonyl)propylzinc bromide, TETRAHYDROFURAN)  
 Hazard Class 4.3  
 Subsidiary Hazard Class 3  
 Packing Group II

**IMDG/IMO**

UN-No UN3399  
 Proper Shipping Name ORGANOMETALLIC SUBSTANCE, LIQUID, WATER-REACTIVE, FLAMMABLE  
 Technical Shipping Name (3-(Ethoxycarbonyl)propylzinc bromide, TETRAHYDROFURAN)  
 Hazard Class 4.3  
 Subsidiary Hazard Class 3  
 Packing Group II

**IATA**

UN-No UN3399  
 Proper Shipping Name Organometallic substance, liquid, water-reactive, flammable



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|                                |                                                         |
|--------------------------------|---------------------------------------------------------|
| <b>Technical Shipping Name</b> | (3-(Ethoxycarbonyl)propylzinc bromide, TETRAHYDROFURAN) |
| <b>Hazard Class</b>            | 4.3                                                     |
| <b>Subsidiary Hazard Class</b> | 3                                                       |
| <b>Packing Group</b>           | II                                                      |

**Special Precautions for User** No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component       | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-----------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Tetrahydrofuran | X                                                   | X                                       | X    | X     | 203-726-8 | X    | X   | X     | X    | X    | X    | KE-33454 |

**National Regulations****SECTION 16. OTHER INFORMATION**

**Prepared By** Health, Safety and Environmental Department  
**Revision Date** 08-May-2024  
**Revision Summary** New emergency telephone response service provider.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

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**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Physical hazards**

On basis of test data

**Health Hazards**

Calculation method

**Environmental hazards**

Calculation method

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**